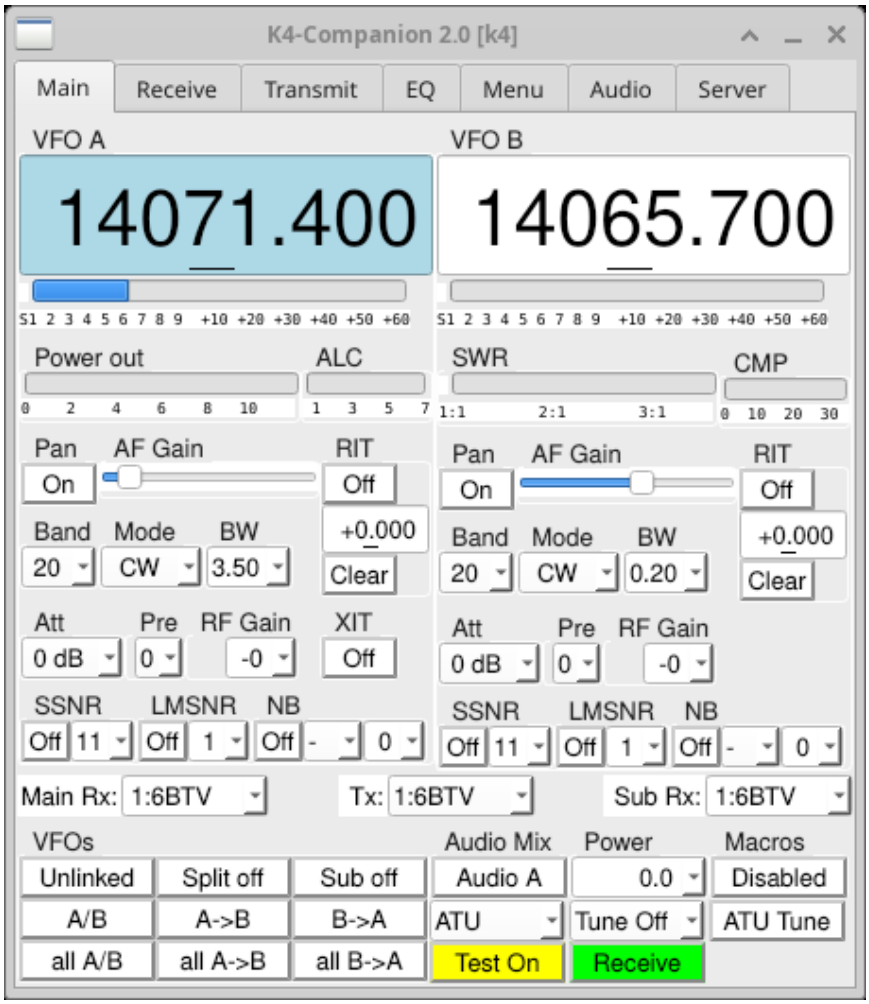
K4 Companion

An application for remote control of the Elecraft K4 series SDR transceiver



Developed by Dale Farnsworth, W7DA

Based on a simple utility originally devised by Charles Powell, NK8O

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The current version of K4 Companion is 2.2

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Introduction

K4 Companion is an application written in python3 that can remotely control an Elecraft K4 transceiver via TCP/IP. It controls the main K4 features, including remote audio, and new features are being added all the time.

Version 2.0 is a major update. The user interface has been rebuilt on Qt (PyQt6), so it looks different on every platform. New in 2.0 are a panadapter and waterfall window for each receiver, and a keyer: K4 Companion can now send CW from your own paddles or straight key connected to the computer, with a sidetone it generates itself. Of course, it can also be used with an external keyer. Connections to the K4 are now encrypted by default.

K4 Companion began life as a simple macro-sending program called K4Macro-Python, created by Charles Powell, NK8O. It has now grown into a full-fledged remote control program for the K4.

Please send problem reports either: by sending an email, by entering an issue on github, or by making a pull request. Problem reports and suggestions are greatly appreciated.

Dale Farnsworth, W7DA
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License

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Windows Installer & Binaries

A Windows installer and Windows binary executable files are available, among the Assets of each release on the K4 Companion releases page, https://github.com/DaleFarnsworth/K4-Companion/releases: the installer, *k4companion-installer.exe*, and the programs it installs, *k4companion.exe* and *k4companion-debug.exe*. **You may need to disable Windows security briefly to download the installer.** After downloading, double click the installer file, then select the desired options. The installer places all files in the installation folder, which by default is the folder *k4companion* in the user's home folder. The executable program, *k4companion.exe*, a version of it with a console for debugging, *k4companion-debug.exe* (see Debugging), and a user manual are installed in the installation folder. A Start Menu entry and a desktop shortcut are also optionally created.

NOTE: *k4companion.exe* carries the Opus audio codec, *opus.dll*, which it needs, within it; the installer places a copy in the installation folder as well. K4 Companion has been tested on both Windows 11 and Windows 10. It is built with Python 3.14.

If preferred, K4 Companion can be run without installing it: download *k4companion.exe* from the release's Assets and put it in a folder of your choice. It needs no other file.

A note regarding Windows Defender: The Defender reaction will range from none at all, to identifying K4 Companion as a virus. If identified as a virus, Defender will quarantine the executable and ask if you want to delete it. Instructions regarding bypassing Windows Defender are found at https://answers.microsoft.com/en-us/windows/forum/all/how-to-allow-an-app-through-windows-defender/caab364c-9e65-448e-8635-98721be847ba

To use K4 Companion from outside your local network, see Remote Access.

Installation on Linux

Installers for Linux are among the Assets of each release on the K4 Companion releases page, https://github.com/DaleFarnsworth/K4-Companion/releases: *k4companion-linux-x86_64.run* for most PCs, and *k4companion-linux-aarch64.run* for 64-bit ARM computers, such as a Raspberry Pi running 64-bit Raspberry Pi OS Trixie. The first needs Ubuntu 22.04, Debian 12 or a distribution of their age or later (glibc 2.35), and the second Ubuntu 24.04, Debian 13 or later (glibc 2.39).

Run the installer with *sh k4companion-linux-x86_64.run* to install K4 Companion for yourself alone, or with *sudo sh k4companion-linux-x86_64.run* to install it for everyone, which also lets a K-Pod be used without root (see K-Pod Tab). K4 Companion then appears in the desktop's menu, and runs as *k4companion* in a terminal. It uses the system's Opus and PortAudio libraries, and the installer says how to install them if they are missing. It also says how to uninstall K4 Companion.

To run K4 Companion from its Python source instead, here are the items one needs to take full advantage of it and all its features:

1. A computer capable of running a full installation of Python is needed. This has been tested with Xubuntu, Debian 12, Chromebook, various iterations of the Dell XPS series, and several Lenovo laptops, and even an Evolve III notebook, running a variety of Linux flavors.
2. Install Python3. This is pre-installed on most Linux distributions. Python 3.13 or later is needed for the encrypted connection on port 9204; with an older Python, use port 9205.
3. Install pip3, required for the remaining dependencies.
4. Install the Opus and PortAudio libraries, which K4 Companion's audio uses, from your distribution's packages. On Debian and Ubuntu: *sudo apt install libopus0 libportaudio2*
5. Install the Python packages with pip: *pip3 install PyQt6 pyqtgraph pyyaml opuslib numpy sounddevice PySocks python-rtmidi pyserial*. For a K-Pod, also install hidapi (see K-Pod Tab).
6. For access from outside your local network, see Remote Access.

Note: recent versions of Debian, Ubuntu and some other distributions refuse to install packages with pip outside a virtual environment, reporting an "externally-managed-environment" error. In that case, create one (on Debian and Ubuntu, install the python3-venv package first) with *python3 -m venv ~/k4companion-venv*, install the packages with *~/k4companion-venv/bin/pip3 install* followed by the package names above, and run K4 Companion with *~/k4companion-venv/bin/python3 k4companion*.

For macOS, see the separate K4 Companion MacOS Installation guide.

Running K4 Companion

There are several ways to run K4 Companion, depending on the operating system. To run the script directly, the file must be marked as executable. Under Linux and Mac OS, this can be done immediately after downloading the script. In a terminal window and using the command line, issue the command *chmod +x k4companion .* No modifications of the Windows executable files are needed.

Methods for starting K4 Companion:

- From a Linux terminal issue the command python3 k4companion, or if the file has been made executable the command *./k4companion*, in the directory where the script is located.
- To start the Windows binary navigate to the file in the chosen folder, then either click on *k4companion.exe* or highlight and press 'e*nter*'
- If the Windows installer has been employed, K4 Companion can be run from the Start Menu or a Desktop icon if either was created at installation time.
- If the Linux installer has been employed, K4 Companion can be run from the desktop's menu, or with the command *k4companion* in a terminal.
- For access from outside your local network, see Remote Access.

Note for Linux users: If you create a shortcut or desktop launcher for K4 Companion, give the full path to k4companion in it, since a launcher does not start in the folder that holds it.

Remote Access

On your local network, K4 Companion connects to the K4 directly. To use it from anywhere else, the K4 must be reachable from the Internet, which can be done in several ways:

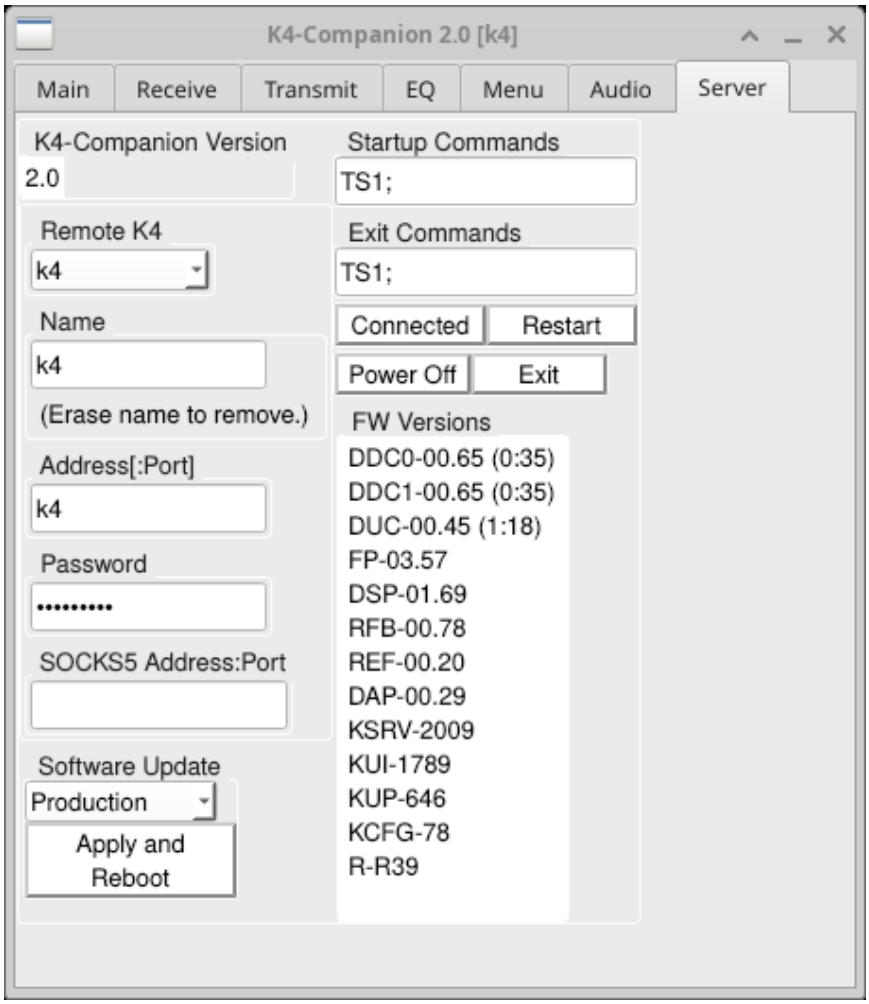
- **Port forwarding.** Forward port 9204 on your router to the K4, and enter your external IP address, or a name you have created with a service such as DynDNS, as the K4's address on the Server tab. If your router forwards a different port to the K4, add that port after the address. When operating from a single remote address, it is possible to "open" your router to accept only that address.
- **VPN.** A VPN, such as Tailscale (https://tailscale.com) or WireGuard, gives access to your internal network without forwarding ports.
- **SOCKS 5 proxy.** Enter the proxy's address on the Server tab. A proxy connection offers greater flexibility in connection options than opening the router to a single address.
- **SSH tunnel.** An SSH 'tunnel' to the K4 can also be used.

Since version 2.0, K4 Companion connects by default on port 9204, the K4's encrypted (TLS) port, which K4 to K4 connections also use. Earlier versions used port 9205, which is not encrypted; it can still be used by adding :9205 to the K4's address on the Server tab, and is then the port to forward.

Remote connections must be enabled on the K4, with a password and the number of remote connections allowed (see Configuration: Server Tab). K4 Companion and another K4 can be connected at the same time if the K4 allows more than one remote connection.

If you are not present at your K4, a way to remotely turn on your K4 is also needed. Please refer to the K4 User Manual or any of several forums that address this function.

Configuration: Server Tab



Server configuration is done through the 'Server' tab. Configuration parameters **must** be entered once but will be saved for future use. The required items are:

- **Specifying the 'Name' for the profile.** Each new profile is given a default name, but it can be changed when the profile is created. Up to 25 profiles are possible with K4 Companion. To add one, choose *new* in the 'Remote K4' pull-down; to remove one, erase its name. In the 'Remote K4' pull-down after a profile is created, select the name for the connection. If only a single connection is created, K4 Companion will connect automatically. Otherwise the program opens to this page so the profile can be selected.
- **The address of the K4 to be controlled.** This can either be the serial number of the radio in the form K4-SNxxxxx.local, or it can be the actual IP address of the K4. There are reports that using the actual IP address makes the connection more responsive, and the serial number setting may not work under Windows. Your experience may be different. For remote operation, see Remote Access. The default is port 9204, which is encrypted. To use the K4's unencrypted port instead, add :9205 after the address.

- **The password to access your K4.** For all remote control this **must** be set on the radio. The default password will NOT work. In addition, the number of connections to your K4 must also be specified. This can be set to one, and up to four connections allowed. No connections will be permitted if the parameter is set to zero or left at default.
- **SOCKS 5 proxy** An optional SOCKS 5 proxy is supported. The SOCKS 5 proxy address is entered in the 'SOCKS 5 Address:Port'. The default port is 1080 and need not be specified unless a different port is used.
- Once information is entered, K4 Companion will attempt to connect. If the information is entered correctly and the server is running, a connection will be established. **Hint:** enter password information first, then the server address
- The Startup and Exit commands are sent to the K4 on startup and exit respectively. By default, they put the radio in Test Mode when K4 Companion opens and return it to Test Mode for safety when the program exits. These can be changed at your own risk!

Firmware upgrades or downgrades, and beta releases, can be installed remotely. Choose the release in the Software Update pull-down (Production, Beta, or Previous Prod., the previous production release), then click Apply and Reboot to have the K4 restart and install it. The FW Versions list on the right shows the firmware installed in the K4, not the releases available. Please check on the Elecraft site for information on the various releases and features, or other information.

On the right hand side, the connection status is shown. 'Power Off' turns off the K4, and the connection status then shows 'Disconnected'. 'Exit' will close K4 Companion without shutting down the radio. 'Restart' can be used if configuration is changed or if there is a brief loss of connectivity, i.e., the Internet connection drops briefly or network congestion causes a problem. 'Connected' reflects the connection status. If the connection to the K4 is lost, K4 Companion says why and tries to connect again. When the K4 reports an ADC overflow, or its COR relay switching, K4 Companion shows it in a warning that goes by itself after a few seconds.

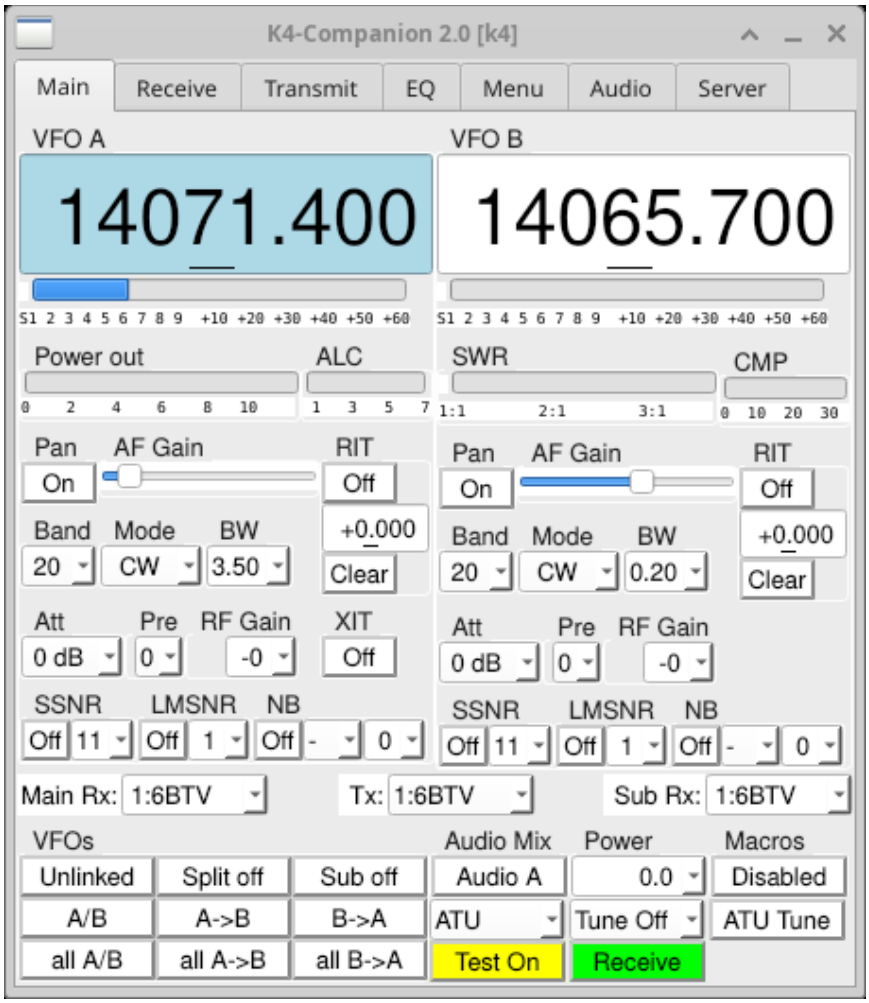
While connected to a K4, the K4's firmware versions are displayed in the lower right.

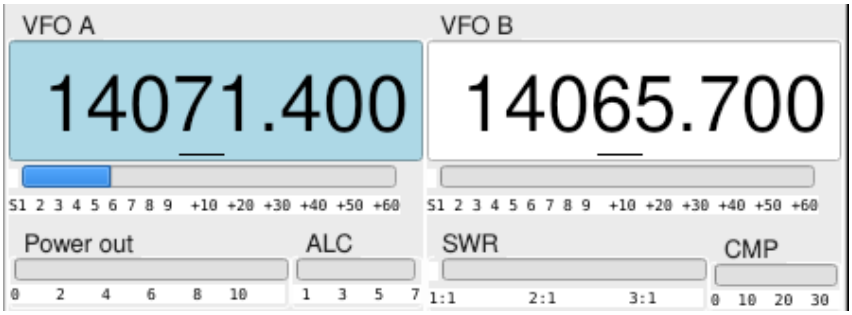
The version of K4 Companion is displayed at the upper left.

Operation and Controls

The operation of many of K4 Companion controls is self-explanatory. Each VFO can be linked, operated independently, swapped, placed into Split mode, Sub-receiver mode, or both simultaneously. A diversity receive selection is also available. Various other functions are controlled in the main tab, including preamps, attenuators, RF gain, ATU functions, and audio controls.

The VFO that will transmit is shown in light blue, so it is possible to see at a glance which VFO will transmit when the radio is keyed. This avoids confusion when using Split or Sub modes.





In the image above, VFO A will transmit. If Split were on, VFO B would be light blue instead. The details available at a glance are the frequency of each VFO, the S-meter reading for each VFO (if both are active), power out either 0-10 watts or 0-110 watts, SWR, and for phone operation, compression. When transmitting, the active transmit VFO turns dark blue.

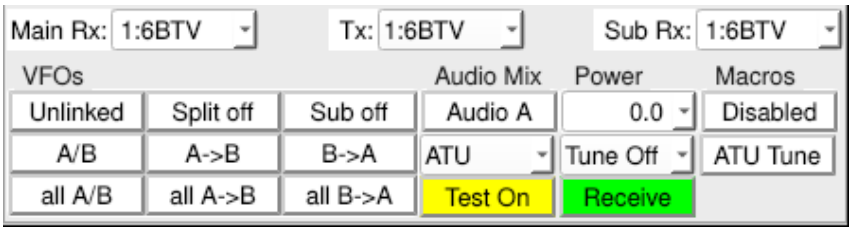
The VFO boxes also support **direct entry.** The increment can be changed between 1 Hz and 10 kHz with the keyboard arrows ← → or clicking with the mouse or touchpad. The active scroll rate increment is underlined. The Up ↕ Down arrows on the keyboard or the mouse wheel can be used to change the frequency by the current increment while the VFO frequency is selected.

Pull-down menus shown in the images above are available for each VFO to set the following parameters:

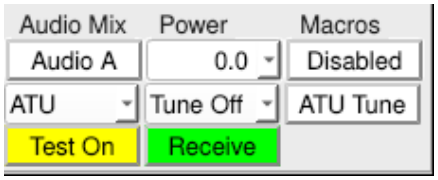
- Band selection, 160 through 6 meters
- Mode selection, including all available modes for the K4 (does not include digital sub-modes)
- Bandwidth selection from 50 Hz to 5 kHz (for AM reception)
- RF attenuation in 3 dB steps up to -21 dB
- Preamp selection − none, 1, 2, or 3 (depending on the band)
- RF gain, in -10 dB increments down to -60 dB. Note: Elecraft recommends use of Attenuation rather than RF gain. Attenuation reduces the signal to the ADC. This is different from the K3
- CMP, beside VFO B's RF gain and under its RIT Clear button − the transmit compression, 0 to 30, as on the Transmit tab
- NB (noise blanker), SSNR and LMSNR (noise reduction), each with an On/Off button and a level. Selecting C in the SSNR level turns SSNR on and resets its noise estimate. AC does the same on the quietest frequency in the panadapter's span, so that the noise is learnt rather than the signal being listened to, then returns to the original frequency.
- Pan opens the panadapter window for that receiver (see Panadapter Window)

Each of these selections will "scroll" with the mouse wheel or up/down arrow keys, in the same manner as the VFO controls.

The following controls are fairly self-explanatory, at least once you are familiar with K4 Companion.



When the application opens, by default it sets the K4 to Test Mode, as shown above. This button is yellow when Test Mode is on. It is also possible to check ALC and compression without actually transmitting.

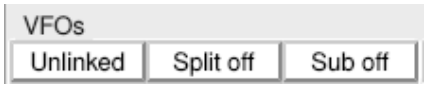


The Transmit/Receive button can be clicked to toggle between transmit and receive and displays the current state. It also changes state when the T/R mode is changed by another function. Pressing the space bar will toggle T/R if the focus is on the main window.

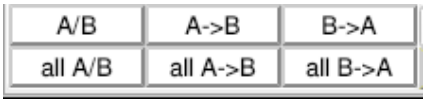
Audio output selection is labeled “Audio Mix”. Available selections are A, B, or A+B. If other selections are specified via macros, these will be reflected in the pull-down, but they are not directly selectable from K4 Companion. XIT is turned on with the XIT button under VFO A’s controls, and its offset is set in the RIT digit box, which XIT shares. The box and the buttons turn green for RIT, blue for XIT, and turquoise when both are on. Both RIT and XIT frequency delta can be selected directly or by scrolling with a mouse wheel or keyboard up/down arrows. Transmit power can be selected either with pull-down presets or by direct entry. The Macros button opens the Macros window (Enabled) or closes it (Disabled); see Macros Window.

In the next row of buttons the options are:

- Linked VFOs or Unlinked. Elecraft calls this “Band Independence”



- Split on, Split off. This button turns split mode on or off; it turns yellow when split is on.
- Sub activates the K4 sub-receiver. This can be used either with Split mode to move the transmit VFO to B, or without. The main and sub-receivers have separate volume slider controls, located below the VFOs and meters.



The middle row of the VFO section is self-explanatory.

- Swap VFOs A/B (may be activated repeatedly to maintain different modes, for example)
- VFO A to B
- VFO B to A

The final row “All” swaps or copies all parameters, equivalent to a “double tap” of the buttons on the K4.

Antenna selection is available in the row beneath the VFOs, for the transmitter (Tx) and for the Main and Sub receivers. These combinations mirror the selections available on the K4, and various combinations can be set in Macros (see Macros Window.)



ATU controls function via pull-down:

- ATU = ATU active
- Bypass = ATU bypassed
- Tune = normal ATU function = single tap on the K4
- TUNE = extended tune = double tap on the K4.



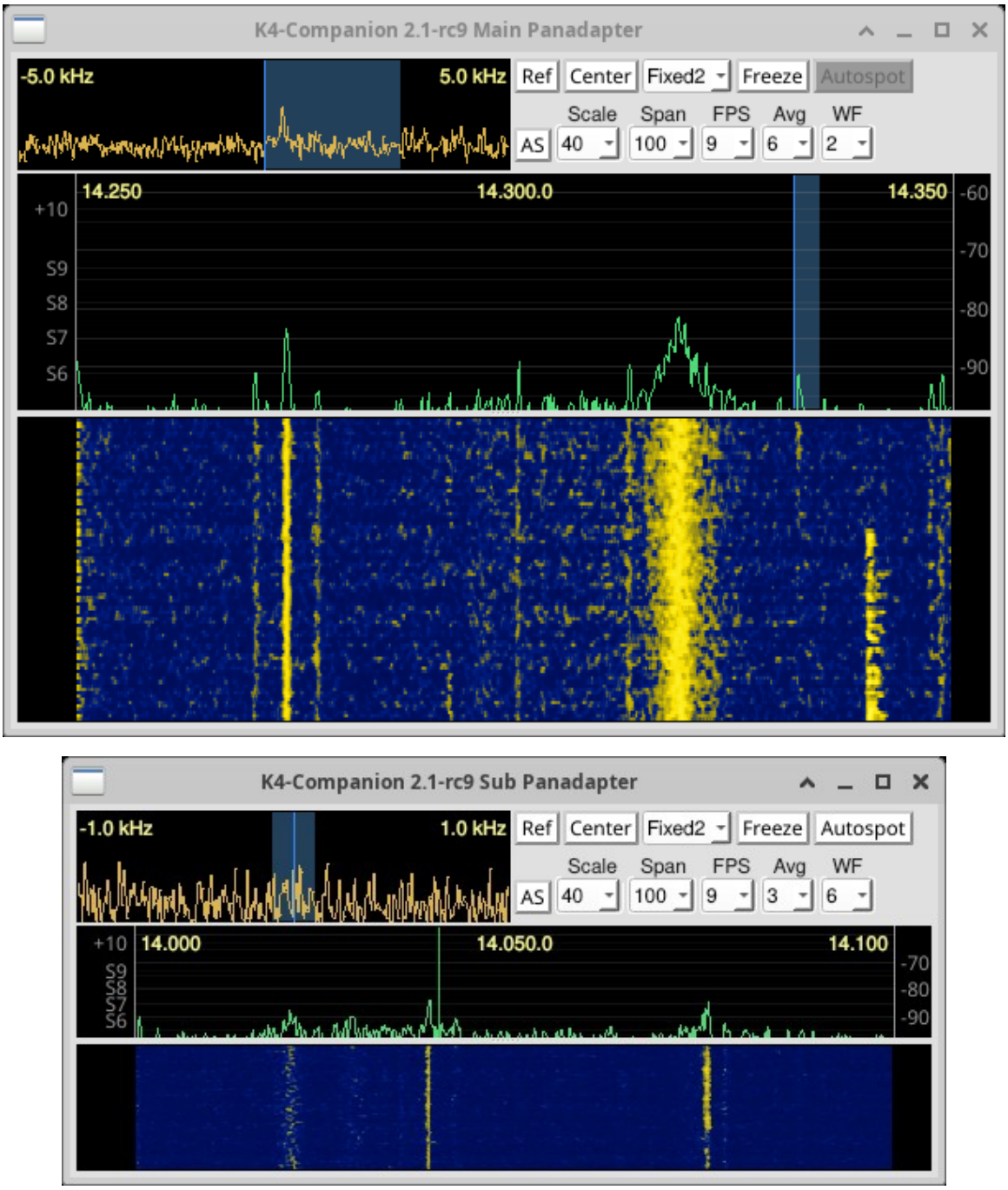
The “Tune Off” pull-down has two additional states:

- Tune produces a carrier at the full (selected) power
- Tune LP produces a low power carrier at 5 watts

The remaining button, “ATU Tune” here, shows the most recently used tune function and that function can be repeated by clicking on this button.

To EXIT K4 Companion, use the window ‘X’ at the top of the frame. Location will depend on your operating system. For Windows and Linux this is typically found at the upper right. On MacOS, it is usually on the left.

Panadapter Window



Each receiver has a Pan button on the Main tab. It opens a panadapter window for that receiver: the K4's spectrum display (the trace) above a waterfall, with a mini-pan, a close-up of the signals around the VFO frequency. The frequencies of both VFOs are marked, and the receive bandwidth is shaded. The waterfall shows signals in yellow on a dark blue background. The left edge of the trace is scaled in S-units, and the right edge in dB. Clicking

Pan again brings an open window to the front. The images show the Main receiver's panadapter at the top and the Sub receiver's below it. In RTTY, a second line marks the space tone, 170 Hz below the VFO in DATA and above it in DATA REV, on the trace, the waterfall and the mini-pan, and the shading is centred between the mark and space tones. The window's position and size are remembered, and it reopens on the monitor it was left on.

Clicking on the trace, the waterfall or the mini-pan tunes the VFO to that frequency. Clicking in the bare strips either side of the waterfall moves half the display span down (on the left) or up (on the right). The left/right arrow keys on the panadapter change the frequency: 1 kHz on the waterfall, 100 Hz on the trace, 10 Hz on the mini-pan and 1 Hz on the button area. The divider between the trace and the waterfall can be dragged.

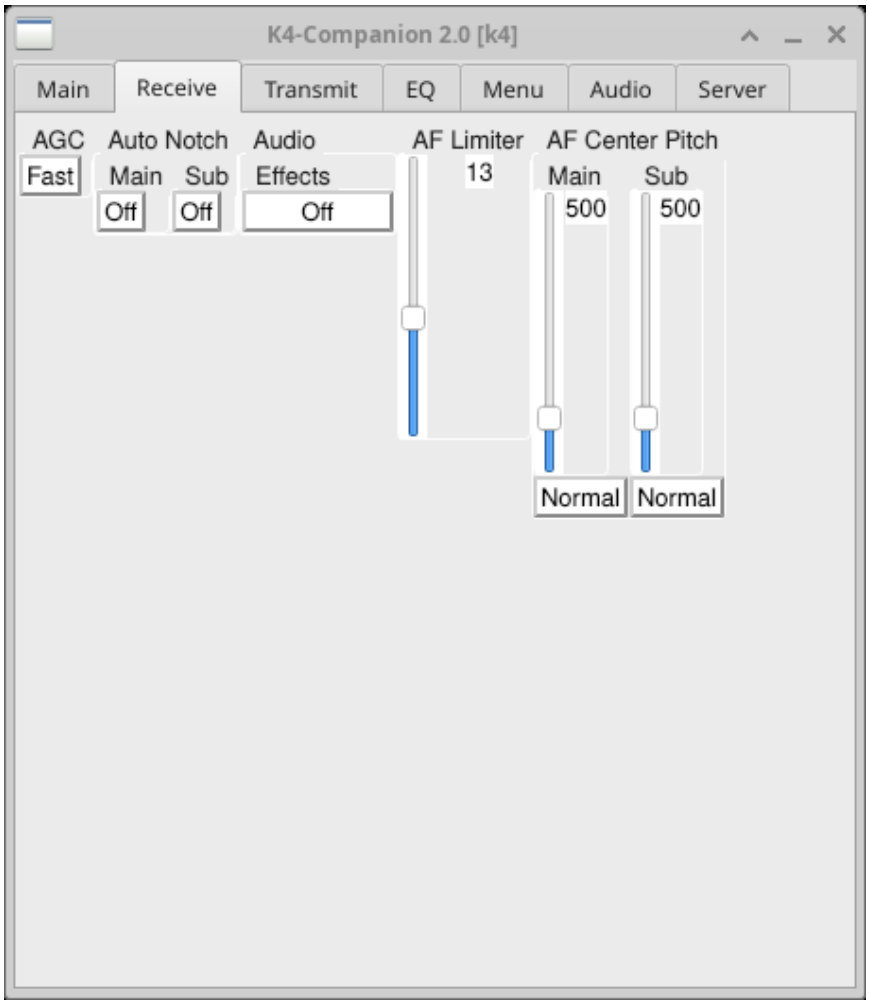
The controls along the top of the window are:

- Span − the width of the display
- Ref − the level at the bottom of the trace, in dB. Auto turns the K4's auto-ref on, which sets it from the noise. Set sets it, for this receiver alone, to the noise floor the trace shows, averaged and offset as the K4's MENU:AutoRef Averaging and AutoRef Offset say, and turns auto-ref off so that it stays there. Or pick a level: those listed run from 7 dB below the noise floor the trace shows to 7 dB above, 1 dB apart, and move with it, so that they suit any band, antenna and span
- Scale − the height of the trace in dB, from Ref up. Set sets it so that the strongest signals of the last few seconds reach about halfway up the trace. Or pick a scale, from 10 to 150 dB in steps of 10
- Center − centers the display on the VFO frequency
- Fixed Tune − Track, Fixed1, Fixed2, Slide1, Slide2 or Static, the K4's fixed-tune modes
- FPS − how many times a second the display is updated
- Avg − how much the trace and mini-pan are smoothed
- WF − how far above the noise floor the trace shows, from 1 to 10 dB, a signal has to be to show in yellow on the waterfall: lower shows weaker signals, higher only the stronger ones. The waterfall's colours follow the noise floor, not Ref, so they stay put while signals come and go
- Freeze / Unfreeze − stops and restarts the display
- Autospot − tunes to the CW signal nearest the VFO (CW and CW-R only)

A Ref or Scale can also be typed into its box, and the mouse wheel or the Up and Down keys over either box step it a dB at a time, as on RF Gain. One between those listed is shown in the box, but not added to the list.

Span, Ref, Scale, FPS, Fixed Tune and Freeze are the K4's own settings, and match the K4's display. Ref is set for each receiver, and its Auto is one for both; the others apply to both panadapters. Please refer to the K4 User Manual for details of each. Scale is remembered for each band of the Main receiver, and put back when the Main receiver goes to that band again; the band the K4 is on when K4 Companion connects keeps the K4's Scale.

Receive Tab

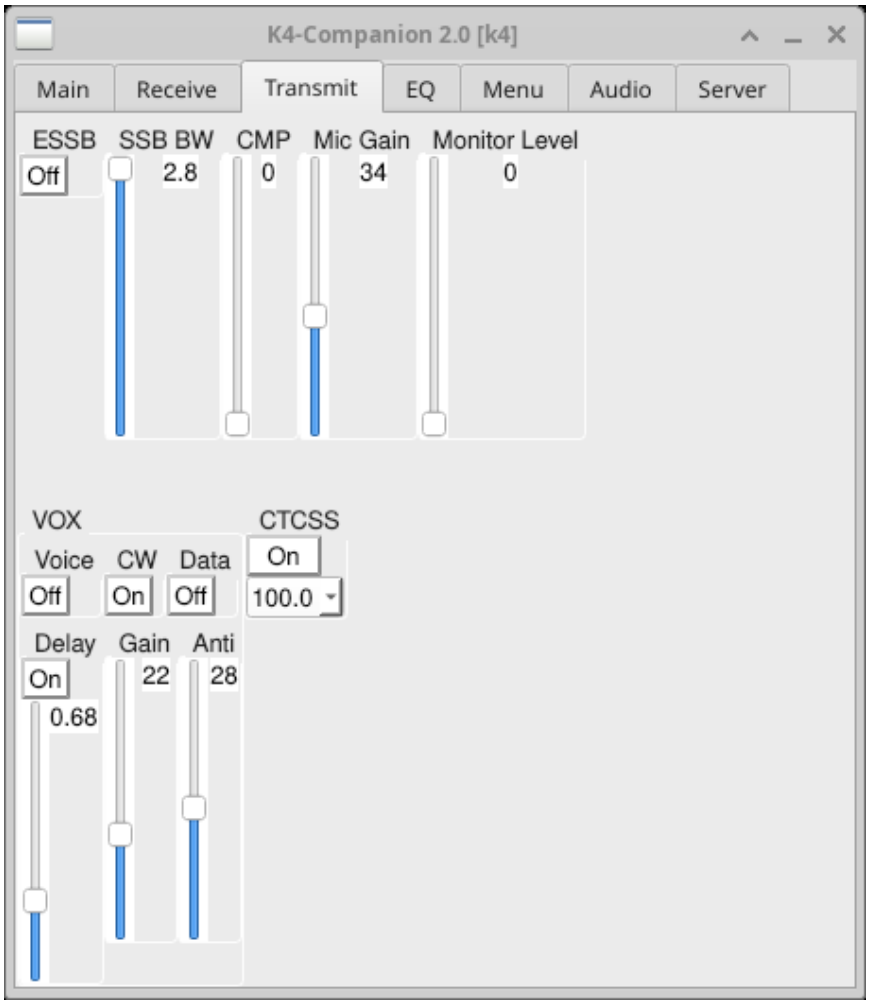


Controls for AGC, Auto Notch (SSB) and Audio Effects are in the top row. The function of each is explained in the K4 User Manual. Options for Audio effects include

- Simulated Stereo – introduces a slight delay in L − R audio to reduce listening fatigue
- Pitch map − moves audio from left to right as the deviation from the center frequency decreases or increases
- Off − removes all tuning and listening effects

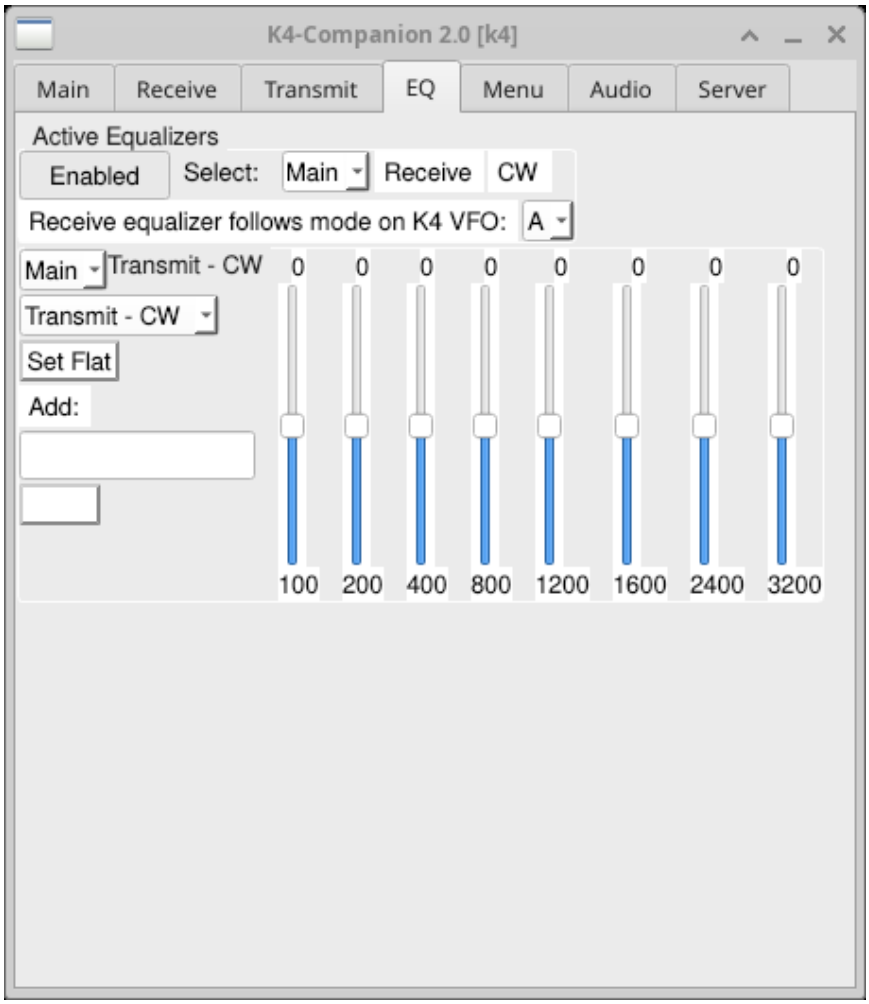
Note that the AF limiter only functions if the AGC is “Off”. This prevents extremely loud signals from blasting through at full-volume if the AGC is absent. AF Center Pitch functions by mode, and returns to the last-used settings when changing modes. The Noise Blanker, SSNR and LMSNR noise reduction controls are now on the Main tab, under each receiver. They are explained in the K4 user manual. Recommended settings vary by band conditions and personal preferences.

Transmit Tab



- Transmit controls include ESSB (wide-band, high fidelity SSB). When selected, the SSB bandwidth varies between 3 kHz and 4.5 kHz
- SSB bandwidth for standard or CESSB (Controlled Envelope SSB) ranges from 2.4 to 2.8 kHz
- CMP − sets the level of transmit audio compression; when set above 0, it automatically enables CESSB. It can also be set from the CMP dropdown on the Main tab
- Mic Gain − set to optimize SSB drive
- Monitor level − useful primarily if using K4 Companion for ancillary control of the radio while operating in the shack. No audio is returned when operating remotely. This control is independent of Sidetone Volume in the CW window.
- VOX on/off with sliders to select parameters
- CTCSS – sets access PL™ tones for FM repeater operation

EQ Tab

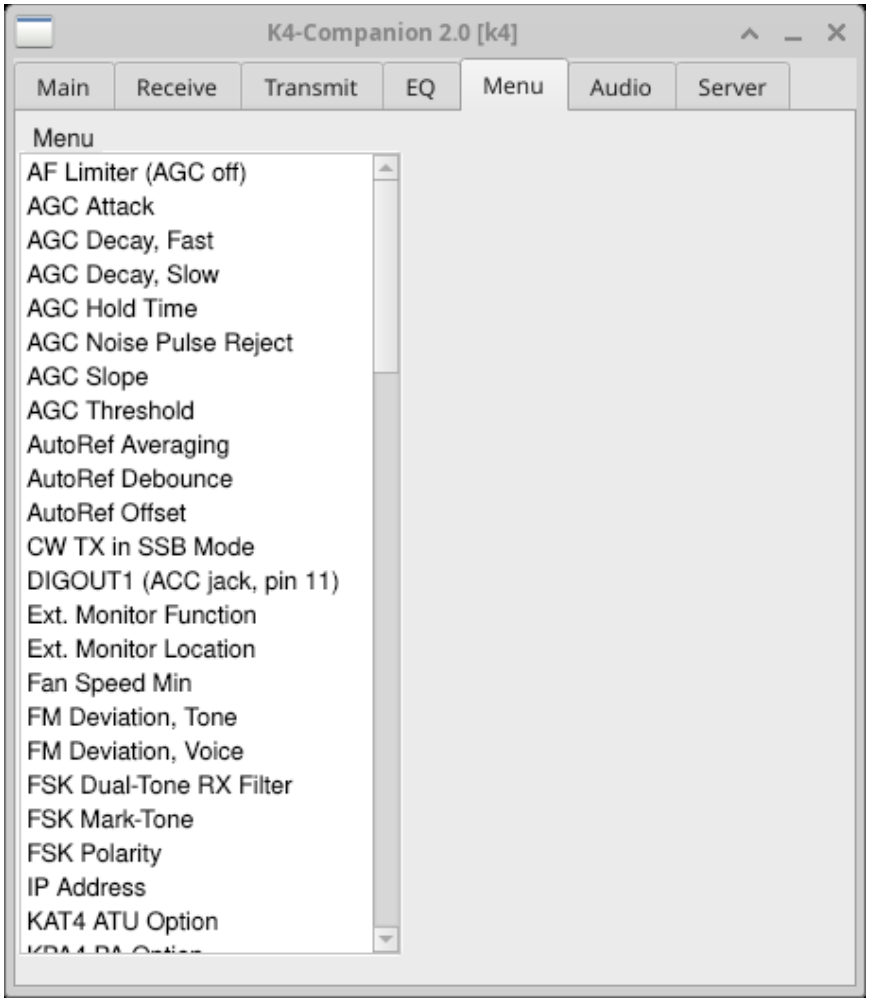


Transmit and Receive equalization can be enabled or disabled, and be set by mode. These are personal preferences. When enabled **the settings on the K4 are altered.** The K4 settings do not revert to previous settings if the Equalizers on K4 Companion are disabled.

Please refer to K4 forums or the K4 User Manual for recommendations.

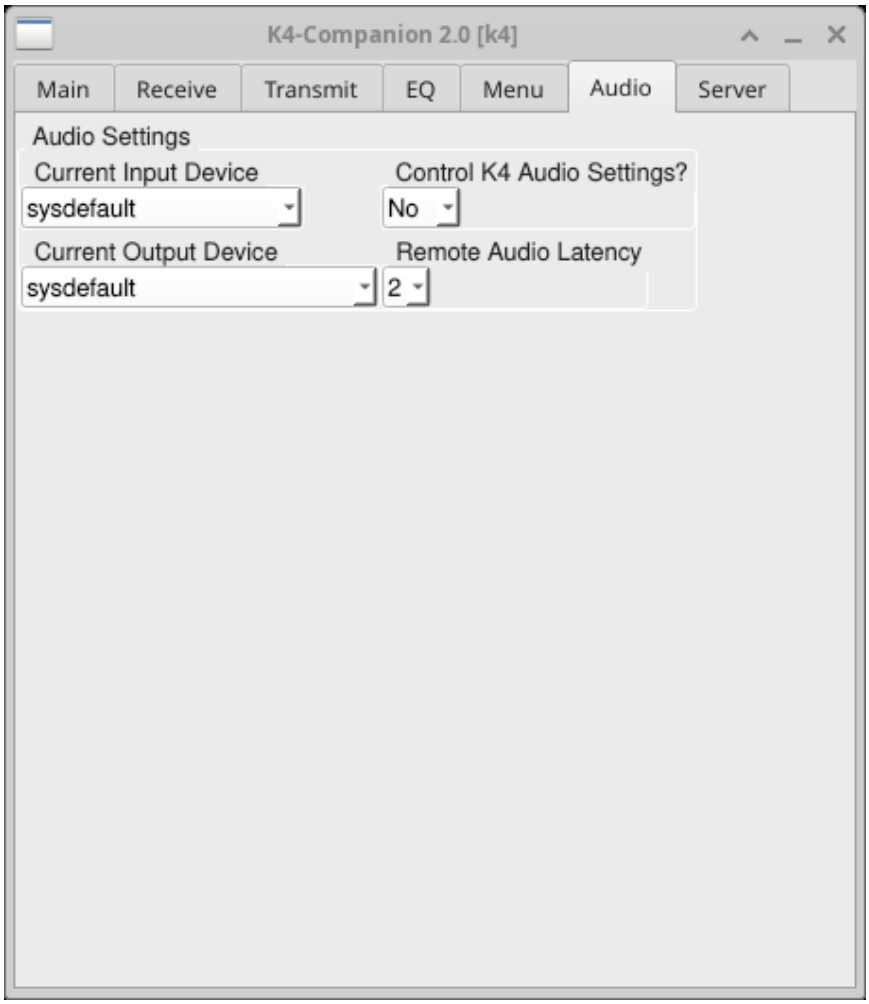
Menu Tab

The Menu tab reproduces **all** of the K4's menu functions. These functions are covered in the user manual. The configuration menu is a long, scroll-able list. When an item is selected, it appears in pink. Then a secondary widget opens that offers the menu choices. Some of these are simple YES/NO selections, others present a list, and some have a slider to select a value for the menu item. Where the K4 gives words instead of a number, such as Ext Ref lock for the reference oscillator while an external reference is locked, they are shown under the slider, which cannot be moved until the K4 gives a number again.



The image shows the start of the menu list. Please be sure to read the K4 user manual carefully. Changing menu settings without due consideration can have unexpected results.

Audio Tab



The Audio tab contains these settings:

- Current Input Device
- Current Output Device
- Exclusive Input Device? and Exclusive Output Device? (Windows only)
- Control K4 Audio Settings?
- Remote Audio Latency
- TCI TX Level %

Typically the Input/Output settings will be correct when K4 Companion opens and they need not be changed. These settings vary by operating system. Audio connections are generally set to ‘default’ in a Linux installation but the choices may vary by OS ‘flavor’, and sound systems installed. In Windows there are a variety of settings, but often the choices mention ‘Speaker’ and ‘Microphone.’ Mac users should consistently see settings that mention ‘Speaker’ and ‘Microphone’ since there is less variability in the hardware.

Each device list also offers None, first. With no output device, neither the K4's audio nor the sidetone is played, though the K4's audio is still decoded for RTTY and SSTV; with no input device, nothing is sent from a microphone. Devices that run at 48 kHz, 96 kHz or 44.1 kHz are listed, and each is used at the first of those it takes, in that order.

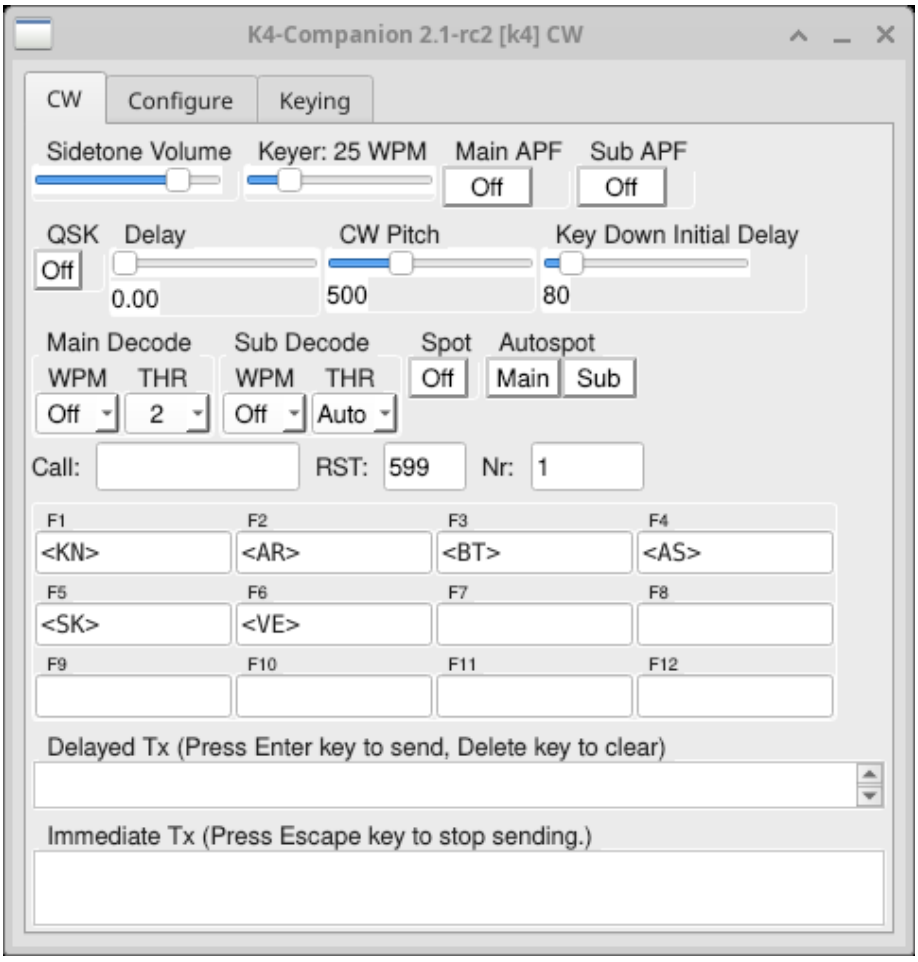
On Windows, each device also has an Exclusive setting. Exclusive mode lets K4 Companion have the device to itself, which lowers the delay of the sidetone when keying from paddles or a straight key. No other program can use the device while K4 Companion has it. The default is No. If a device cannot be opened exclusively, K4 Companion says so and shares it instead.

Control K4 Audio Settings? If you are using K4 Companion to control the radio while sitting in the shack, this can be set to Yes. Otherwise for remote operations, set it to No. See Notes on Operation below. Once this tab is configured, the settings are saved automatically. No changes are needed unless something in your operational configuration changes. The equalizer settings do change on the radio, regardless of the settings selection in the Audio tab. Frequent remote users may want to set the radio's audio to zero, and 'Control K4 Audio Settings?' to 'No'. In this way, the radio will be silent during remote operations.

Remote Audio Latency sets the K4's own latency for the audio it streams, from 0 to 7. It is useful for remote connections where there is considerable jitter on the network path. The lower settings are usually acceptable in the home network setting. A higher setting could be needed for Internet connections with higher variability, if the audio breaks up. Increasing this setting adds additional delay to the received audio.

TCI TX Level % sets how loud a TCI program's transmit audio goes to the K4 (see TCI for Digital-Mode Programs).

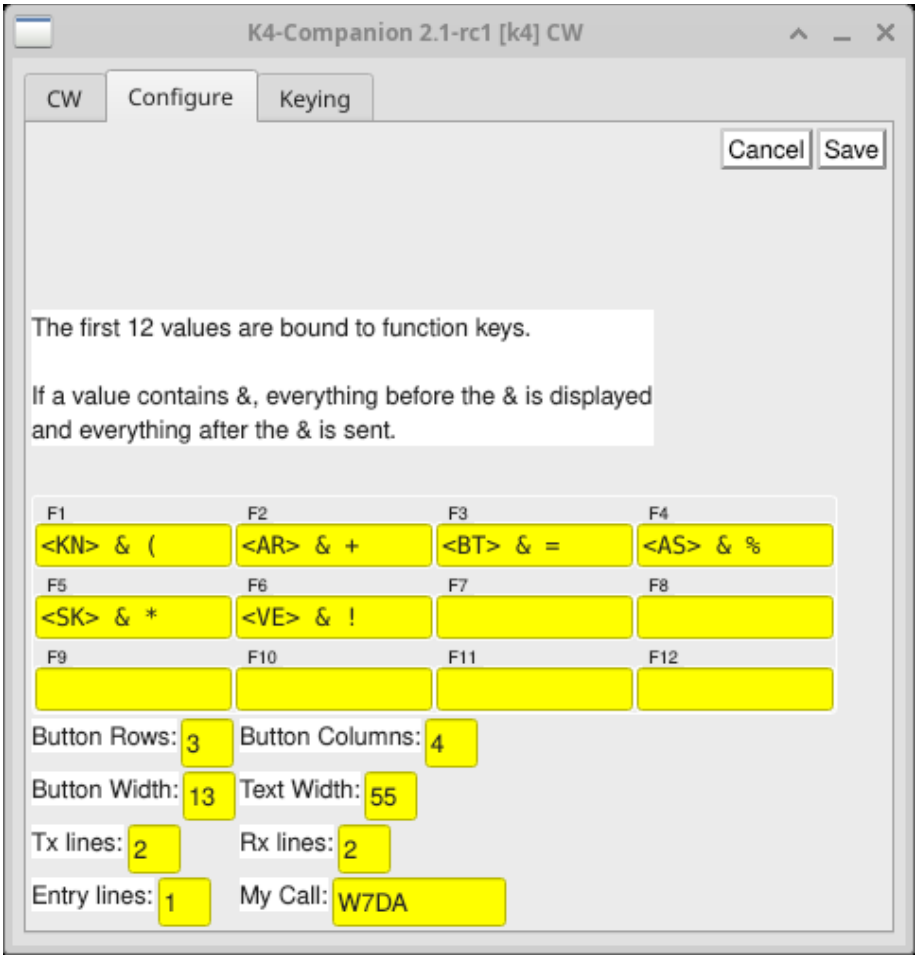
CW Window



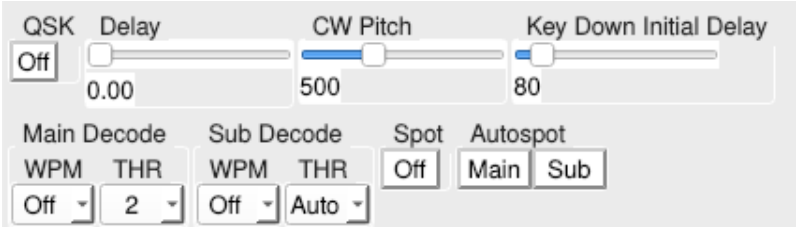
K4 Companion has a CW functionality via the keyboard. There are a number of user-configurable macros that are easily set. These can be used sequentially in any combination. The typing buffer holds any text that is set until the user hits ‘enter’. The text will be sent and shown below in the Tx box. A cursor follows the text that is being sent. The sidetone will also reproduce each character audibly, unless the Sidetone Volume is set to zero. The CW window has three tabs: CW, Configure and Keying (see Keying Tab). Keyer speed and sidetone volume controls are on the upper left of the CW tab. The APF – Audio Peaking Filter − and Autospot each have separate Main and Sub selections. The K4 sends no spot tone to a remote client, so K4 Companion makes its own (see Spot, below).

Several combination items require special characters to send correctly, but they have designations common to Elecraft. For example, ‘AR’ is sent with ‘+’ and ‘SK’ is sent with ‘*’. Other prosigns include ‘(’ for **KN**, ‘=’ for **BT**, ‘%’ for **AS**, ‘_’ for **AA** (used in sending message traffic) and finally ‘!’ for **VE**. To label these items, use the “&” as the divider to differentiate the label from what is actually sent. The label can be anything you like that tells you what is

being sent. Symbols that will not transmit on the K4 are excluded and a brief warning appears if any is entered in the text box.



Editing the buttons is accomplished with the ‘Configure’ tab. The ampersand (&) is used between the label and what is actually sent. To finish editing, click ‘Save’. Changes can be made at any time. In the default configuration there are 20 macro buttons. The number of rows, columns, and other configuration can be edited to taste in the ‘Configure’ mode, but there is no harm in leaving the default configuration. One might use one row for CWT, another for hunting parks, and another for chasing DX. There is no distinction between upper case and lower case letters when sent. Brackets <> can be used in button labels, but are not sent: on the transmit side of a button they enclose the tags described below.



Some of the less obvious Elecraft CW designations are installed by default on the primary setup. These can be moved, changed, deleted, or edited to suit your preferences. Finally, sending can be quickly canceled by hitting 'Esc' − escape.

A button can also send tags, as fldigi's macros can, each put in place when the button is pressed:

- <MYCALL> − your call, set as My Call on the Configure tab, one for both the CW and RTTY windows
- <CALL> − the other station's call, from the Call field above the buttons
- <RST> − the report in the RST field, 599 unless changed
- <CNTR> − the serial number in the Nr field, which is remembered
- <INCR> and <DECR> − add one to the serial number, or take one away, sending nothing
- <CLRLOG> − empties the Call field and puts RST back to 599, ready for the next contact

For example, a contest exchange button might be EXCH & <CALL> 599 <CNTR> <CNTR>, and the one after it TU & TU <MYCALL><INCR><CLRLOG>. A tag that is not one of these is refused when the buttons are saved.

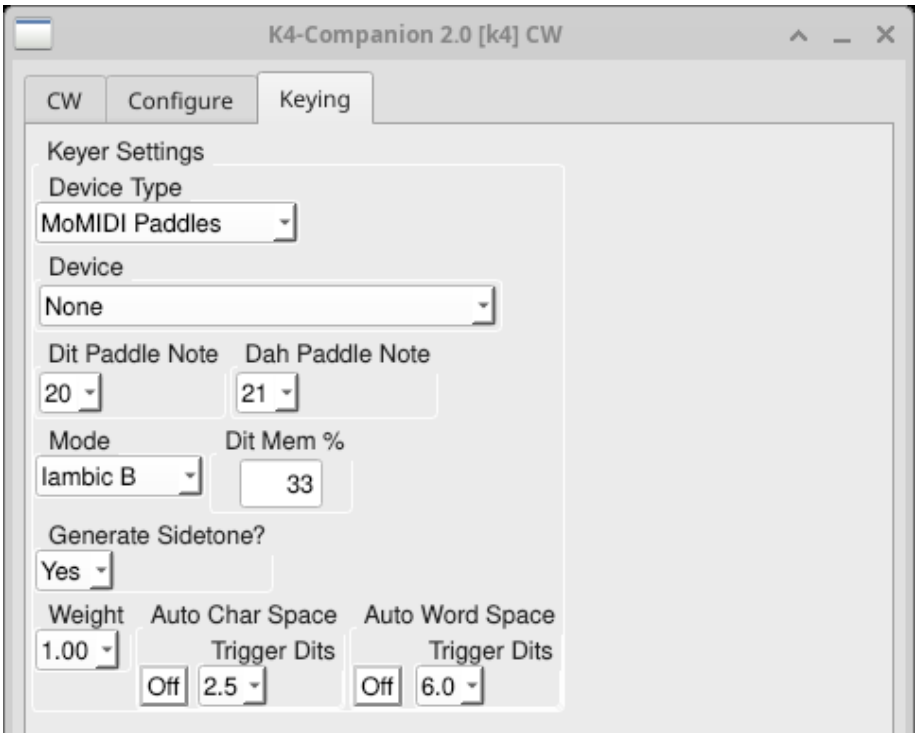
Clicking a call in the Main Rx or Sub Rx text puts it in the Call field. Words that are not calls, 599 or TU say, and your own call are passed over.

QSK on and off, the Delay, CW Pitch and Key Down Initial Delay are also set on the CW tab, as are the Main and Sub CW decoders (WPM and THR).

Spot, beside the decoders, turns on a steady tone at the CW pitch, made by K4 Companion and mixed with the received audio, for tuning a signal to the same pitch by ear, as the K4's SPOT is used. It plays at the Sidetone Volume, is faded away while you send, and turns off when the CW window closes or the K4 leaves CW.

When sending from the keyboard, the sidetone comes from the K4. If it is absent despite the level being turned up, check in the Menu tab of the Main window. "TX Monitor Level, Remote" must be set to zero. If it is set to -1, there will be no sidetone heard on transmit. When keying from paddles or a straight key, K4 Companion generates the sidetone itself (see Keying Tab).

Keying Tab



K4 Companion can key the K4 from paddles or an external keyer or straight key connected to the computer. Its keyer forms the dits and dahs, sends them to the K4, and plays a sidetone on the computer, so that you hear your sending without waiting for the audio to come back from the K4. The Keying tab of the CW window sets it up. The keyer works only when the K4 is in CW mode, and it uses the keyer speed set on the CW tab.

Device Type selects how the key is connected:

- None − no key is connected
- MIDI Paddles or MIDI Straight Key − a MIDI keying interface. Choose the MIDI device, and the note sent by each paddle (Dit Paddle Note and Dah Paddle Note) or by the key. Examples of devices supporting this interface are the HaliKey Midi, electronics.halibut.com/product/halikey/, and the CTR2 series, ctr2.lynovation.com/category/products/.
- Serial Paddles or Serial Straight Key − paddles or a key wired to a serial port. Choose the port, and the signal (DSR, DCD, CTS or RI) each paddle or the key is on
- Vail Adapter − a Vail adapter, www.vailadapter.com, chosen from the MIDI devices. Its own keyer modes are offered in the Vail Mode list. In all but Passthrough the adapter keys itself, and K4 Companion sends what it keys to the K4 as from a straight key. In Passthrough the adapter sends the paddles as they are, and K4 Companion's own keyer keys them, in the mode chosen in the Mode list that appears under Vail Mode

An external keyer is seen by K4 Companion as a straight key, so that is the setting to be used for external keyers.

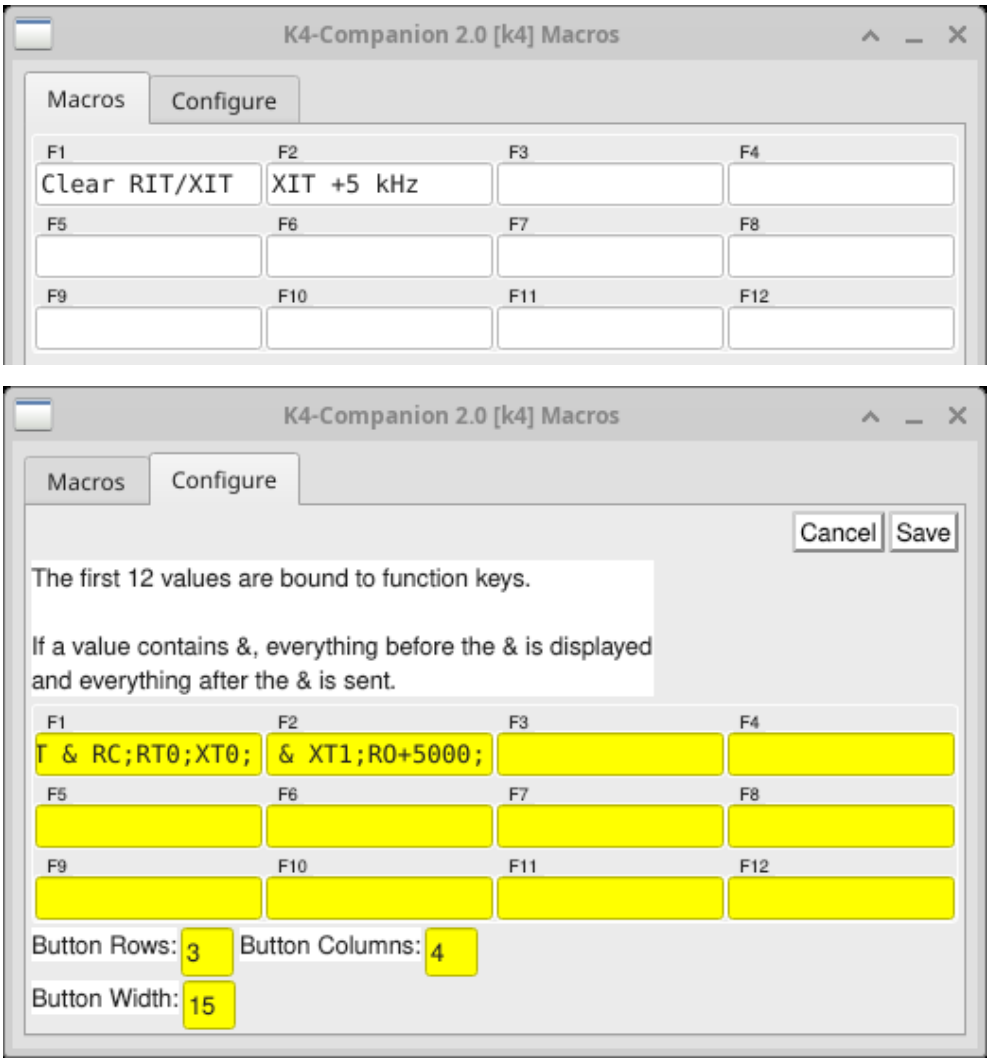
Mode (paddles, and a Vail adapter in Passthrough) selects Plain Iambic, Iambic A, Iambic B or Ultimatic. In Iambic B, Dit Mem % is the leading percentage of a dah in which releasing a squeeze sends no dit after it. Hovering over it gives a reminder.

Generate Sidetone? chooses whether K4 Companion plays the sidetone itself. Yes is recommended unless using a device with its own sidetone. Its volume follows Sidetone Volume on the CW tab. While you are keying, K4 Companion mutes the K4's own monitor sidetone, so that it is not heard as well, and restores it afterward. The sidetone is heard a little after the paddle closes, by the delay of the computer's audio output; on Windows, exclusive mode on the Audio tab shortens it.

Weight, from 0.90 to 1.25, lengthens each dit and dah (above 1.00) or shortens it (below 1.00), taking the difference from the space after it, so the speed is unchanged. 1.00 is standard.

Auto Char Space and Auto Word Space, when On, lengthen a pause to a full character space or word space. Trigger Dits is how long a pause, in dits, counts as one: 1.5 to 3.0 for a character space (2.5 by default), and 5.0 to 7.0 for a word space (6.0 by default).

Macros Window



Macros add great flexibility and control to the K4. Each button in the Macros window sends a string of K4 commands, and its label is shown briefly when it is clicked. The first 12 buttons are also bound to the function keys F1 to F12, which work when either the Main window or the Macros window has the focus.

The Macros button on the Main tab opens the window (Enabled) or closes it (Disabled), and K4 Companion remembers which. The window can also be closed by clicking the X in its corner.

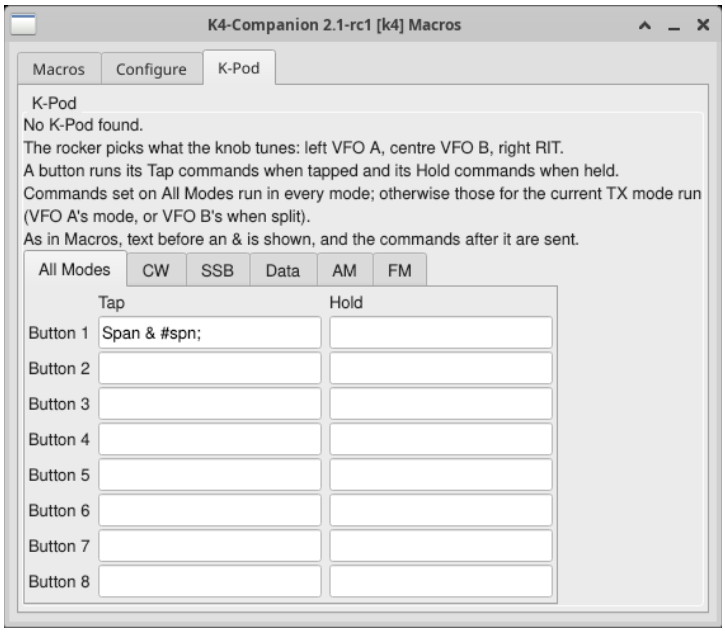
The buttons are edited on the Configure tab. Each field holds a button's label and the commands it sends, separated by an ampersand (&) in the same way as in the CW window: everything before the & is shown on the button, and everything after it is sent. A field with no

& shows and sends the same text, and a semicolon is added to the end of the commands if it is missing. Click Save to keep the changes, or Cancel to discard them.

Button Rows and Button Columns, each from 1 to 20, set the size of the grid, 3 rows of 4 buttons by default, and Button Width sets how wide each button is.

Please refer to the Elecraft K4 Programmer’s Reference or various K4 user forums for more information regarding macro strings.

K-Pod Tab



K4 Companion supports the Elecraft K-Pod, the USB control pod with a tuning knob, a three-position rocker and eight buttons, plugged into the computer. It must be plugged into the computer K4 Companion runs on, not into the K4: a K-Pod plugged into the K4 works the K4 directly, and K4 Companion cannot see it. It can be plugged in or out at any time: K4 Companion looks for it every few seconds.

The rocker picks what the knob tunes: left VFO A, centre VFO B, and right the RIT offset. Each click of the knob moves it one step, at the step the VFO or RIT is set to.

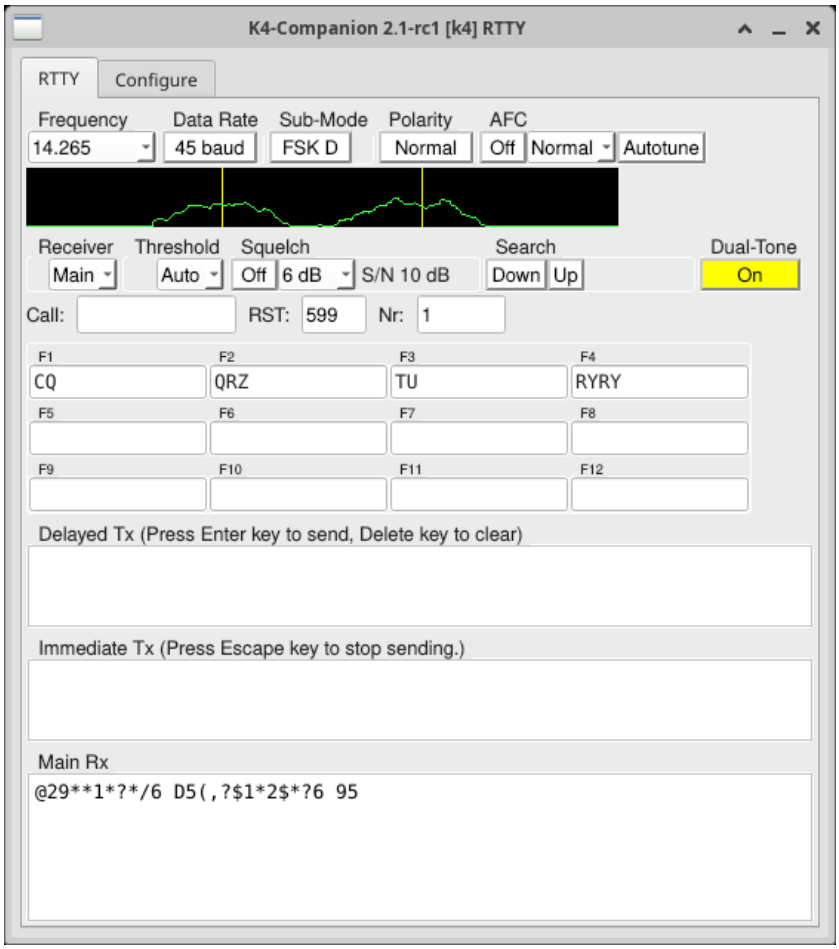
Each button runs K4 commands, set on the K-Pod tab of the Macros window: one set when the button is tapped, and another when it is held. As in the Macros window, the text before an & is what is shown, and the commands after it are what is sent. The commands are kept with each K4's settings, like macros. The top of the tab says whether a K-Pod is connected and what the rocker is set to.

The tab has a sub-tab of buttons for All Modes, and one each for CW, SSB, Data, AM and FM. A button's All Modes commands run in every mode, and while it has any, that button is greyed out on the other sub-tabs. Otherwise the commands for the transmitting VFO's mode run: VFO A's, or VFO B's when split.

The K-Pod needs the hidapi Python package, which the Windows installer includes. On Linux, install it with *pip3 install hidapi*, or as python3-hid from Debian's or Ubuntu's packages; without it, the K-Pod tab says what to install. On Linux the K-Pod also has to be accessible to you: if the tab says a K-Pod was found but could not be opened, create the file */etc/udev/rules.d/70-kpod.rules* holding the line *SUBSYSTEM=="hidraw", ATTRS{idVendor}*

=="04d8", ATTRS{idProduct}=="f12d", TAG+="uaccess" and then unplug the K-Pod and plug it in again.

RTTY Window



K4 Companion has an RTTY window, patterned after the CW window, for RTTY: the K4 decodes it and sends it in FSK D, K4 Companion sends it in AFSK A, and the window gives it a keyboard, macro buttons and controls. It opens whenever the transmitting VFO is in DATA or DATA REV with the AFSK A or FSK D data sub-mode. RTTY in a VFO's mode dropdown on the Main tab puts that VFO in DATA with FSK D, and the filter last used at its data rate, and turns compression off (CMP 0).

In FSK D the K4 sends the text itself. In AFSK A, where it cannot, K4 Companion makes the RTTY audio and sends it to the K4 in place of the microphone's: the K4 is put in transmit when there is text to send, LTRS are sent while there is none, and it is put back to receive 2 s after the last of it, or at once with Escape. How loud it is sent is AFSK Level % on the Configure tab, 40 unless changed: set it so that the K4's ALC stays just below its 5 mark while sending, as for other audio data.

Typing, sending and the macro buttons work as in the CW window: text typed in Delayed Tx is sent when Enter is pressed, text typed in Immediate Tx is sent as it is typed, and Escape stops sending at once. The tags, the Call, RST and Nr fields, and clicking a call in the

received text work as there too, My Call being one for both windows. The buttons are edited on the Configure tab, with an & between a button's label and what it sends. Only what RTTY can send can be typed: letters, figures and Baudot's punctuation, less the semicolon. Sending is paced by the data rate. There is no keyer, so there is no Keying tab. Decoded text from the main and sub receivers is shown below. The RTTY window's buttons and settings are kept apart from the CW window's.

The controls along the top act on VFO A, which the K4 transmits on unless split:

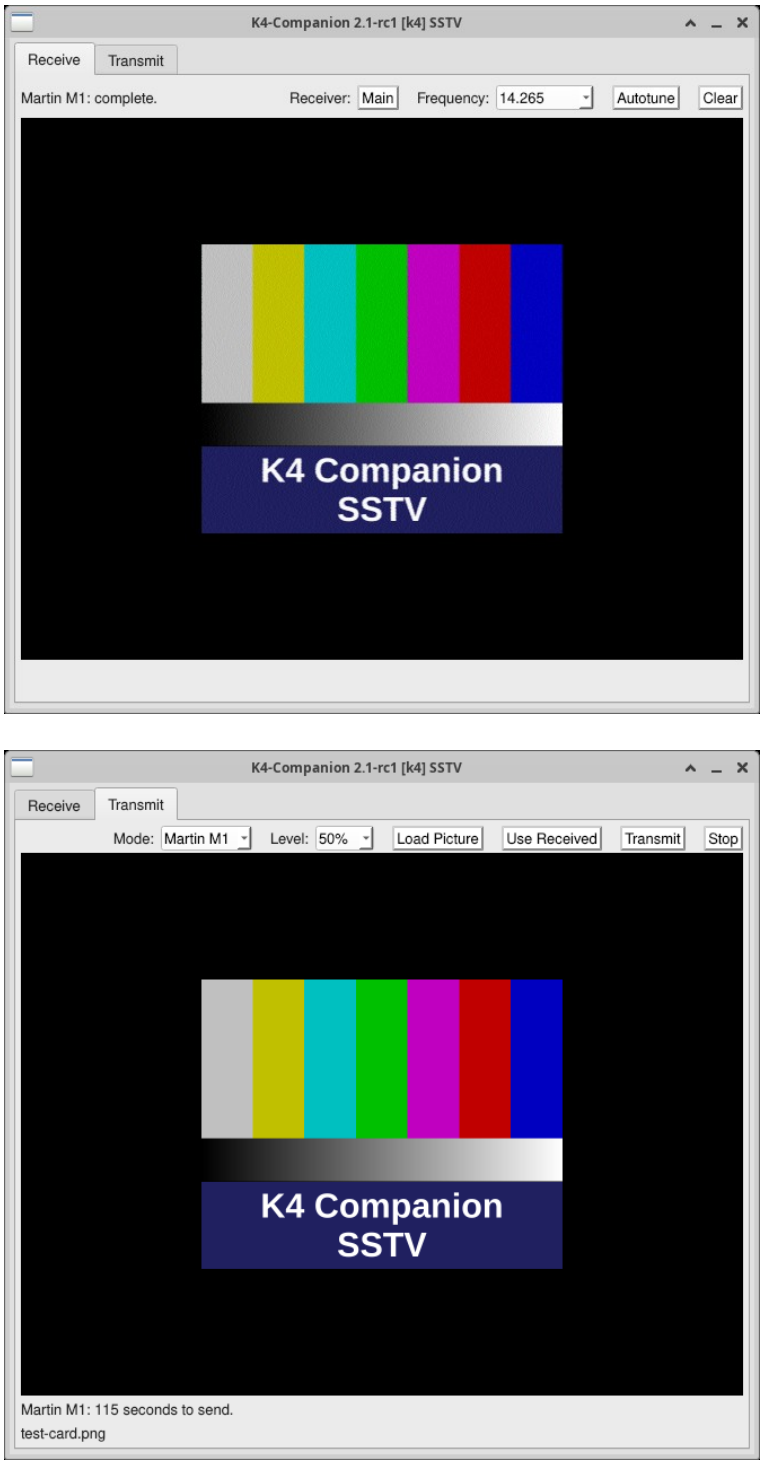
- Frequency − where RTTY usually is on 80, 40 (Region 1 and Region 2), 30, 20, 17, 15, 12 and 10 m. Picking one tunes VFO A there, in RTTY's mode and sub-mode. The dropdown shows VFO A's frequency, to the nearest kilohertz, whether or not it is one of them
- Data Rate − 45 or 75 baud. The filter is remembered for each rate, starting at 300 Hz for 45 baud and 400 Hz for 75, and put back when the rate changes
- Sub-Mode − AFSK A or FSK D
- Polarity − Normal (DATA) or Reverse (DATA REV)
- AFC − when on, keeps RIT where the RTTY signal coming in sits on the K4's mark and space tones, checking every 0.6 s. Beside it, Slow, Normal or Fast sets how quickly it follows: Slow rides through a signal's wobble, and Fast catches a drifting one sooner. Autotune moves VFO A to the signal instead, folding in RIT's offset and setting RIT to zero, so that what is sent goes where the other station is

Under them, a strip shows the receive audio around the K4's mark and space tones, which two lines mark: an RTTY signal tuned in shows a peak on each line.

Below it:

- Receiver − which receivers decode: Main, Sub or Both. At least one always does
- Threshold − the K4's decode threshold, Auto or 1 to 9, set for both receivers
- Squelch − when on, decoded text is shown only while an RTTY signal is coming in on that receiver, at least the level picked beside it above the noise, so that noise, a carrier or a voice does not fill the window with garbage. It opens when a signal is found, and stays open for 2 s after. S/N shows how far the strongest RTTY signal near the K4's tones is above the noise, as a guide to the level: 6 dB, the default, keeps out nearly all noise, and a lower level lets weaker signals through along with more of it. While the squelch is on, AFC follows only signals above its level
- Search − Down and Up tune VFO A to the next RTTY signal below or above, found in the K4's own spectrum, and set RIT to zero. With the main panadapter's span narrow enough to show an RTTY signal's two tones apart, they look across the whole span; otherwise they look in the K4's mini-pan, which reaches about 2 kHz above the tones but only about 150 Hz below them. Beside them is shown for a few seconds how far they moved, None found, or Wait: they need a second of the spectrum after the window opens or after a move
- Dual-Tone − the K4's FSK Dual-Tone RX Filter: separate mark and space filters, which may improve copy. The K4 has it only with its 915 Hz mark tone

SSTV Window



K4 Companion can receive and send slow-scan television (SSTV) pictures. The K4 has no SSTV of its own, so K4 Companion decodes the pictures from the receive audio, and makes

the audio for those it sends. Martin M1 and M2, Scottie S1, S2 and DX, Robot 36 and 72, and PD 50, 90, 120, 160, 180, 240 and 290 can be received and sent.

SSTV in a VFO's mode dropdown on the Main tab turns it on for that receiver. The VFO is put in the sideband SSTV uses there, USB from 14 MHz up and LSB below, and kept in it as it is tuned across, and its filter is set to 2.8 kHz. Compression is turned off (CMP 0), as it would distort the picture's tones. The SSTV window opens, with a Receive tab and a Transmit tab. Picking USB or LSB from the dropdown, putting the VFO in another mode, or closing the window turns SSTV off. K4 Companion remembers it, and turns it back on at the next start if the VFO is still in USB or LSB.

The Receive tab shows each picture as it comes in, recognising its mode from the code sent at its start. When a picture ends, it is saved as a PNG file in the K4-Companion SSTV folder in your Pictures folder, once a tenth of it or more has come in. The status line says which sideband is in use, or the mode and how far a picture has got. Its controls:

- Receiver − Main or Sub: which receiver SSTV is on, moving it as that VFO's mode dropdown would
- Frequency − the middle of where SSTV usually is on 80 m (3.845 MHz, or 3.730 in Region 1), 40 m (7.171), 20 m (14.230), 15 m (21.340), 10 m (28.680) and 6 m (50.680). Picking one tunes the VFO there, in the right sideband with the 2.8 kHz filter
- Autotune − moves the VFO so that the signal coming in sits where its tones belong, from where its sync pulses are in the last few seconds of audio. RIT's offset, if RIT is on, is folded into the VFO and RIT set to zero. What it did is shown in the status line for a few seconds
- Clear − clears the picture

The Transmit tab sends a picture:

- Frequency − as on the Receive tab, the two showing the same; it cannot be changed while a picture is being sent
- Load Picture − picks a picture file: PNG, JPEG, BMP or GIF. The last one loaded is loaded again at the next start
- Use Received − picks the picture last received, to send back
- Mode − the SSTV mode to send in. The status line says how long it takes
- Level − how loud the picture is sent, from 5% to 100%, 50% unless changed. Set it so that the K4's ALC just shows while sending, as for voice
- Transmit − puts the K4 in transmit, sends the picture, and puts it back to receive at the end. The status line shows how far it has got
- Stop − stops sending, and puts the K4 back to receive

The picture is scaled to fill the mode's size, and what is left over either side is cut off evenly; the tab shows it as it will be sent. It is sent on the transmitting VFO, VFO A, or VFO B when split, which has to be in USB, LSB, FM or AM. While a picture is being sent, it goes to the K4 in place of the microphone's audio. Turning SSTV off or closing the window while sending stops it and puts the K4 back to receive.

CAT for Other Programs

Logging and digital-mode programs on the same computer can control the K4 through K4 Companion, sharing its connection, rather than connecting to the K4 themselves. K4 Companion accepts their connections on TCP port 9200 of the computer it runs on, as the K4 does on its own port 9200, and passes their commands on to the K4. Set such a program up for an Elecraft K4 connected by network, at address 127.0.0.1 and port 9200; in a program using hamlib, choose the Elecraft K4. This works even where only K4 Companion can reach the radio, as when it connects from outside the K4's network (see Remote Access).

A program's questions are answered with the K4's replies, or, when the K4 has not replied within a second, with the last value K4 Companion has. Its commands change the K4 at once, and K4 Companion's windows follow, as if they had been made in K4 Companion. An error the K4 gives for one of its commands is answered to the program, as the K4 would answer it, rather than shown by K4 Companion.

The commands that set how the K4 talks back, AI and the K2, K3 and K4 extended modes, are not passed on, since they would change it for K4 Companion as well; each program is answered as if they had been.

Only programs on the same computer can connect, since anything that can connect can put the K4 in transmit. If another program on the computer already uses port 9200, K4 Companion cannot accept connections there.

TCI for Digital-Mode Programs

K4 Companion also speaks TCI, the Transceiver Control Interface of Expert Electronics' SunSDR radios, which JTDX, WSJT-X Improved and other digital-mode programs support. Through it a program controls the K4, and is sent the K4's receive audio and sends its own transmit audio over the same connection, so that no audio cables, real or virtual, are needed. K4 Companion accepts TCI connections on port 40001 of the computer it runs on.

To set up JTDX or WSJT-X Improved, in its Settings, on the Radio tab, choose TCI Client RX1 as the rig, enter 127.0.0.1:40001 as the TCI server, choose CAT as the PTT method, and tick TCI audio. The program's audio input and output are then TCI audio.

The main receiver is TCI's receiver 0: its frequency is VFO A's, and its second channel, used when split, is VFO B. The sub receiver is receiver 1. The program can set the frequency, the mode (DIGU and DIGL are the K4's DATA and DATA REV, in its DATA A sub-mode), split, PTT, Tune and RIT, and reads the S-meter and, while transmitting, the power and SWR. K4 Companion's windows follow what it sets, as for CAT.

The K4's receive audio is sent to the program at 48 kHz. While the program transmits with TCI audio, its audio goes to the K4 in place of the microphone's, in any mode, DATA included. How loud is set by TCI TX Level % on the Audio tab, 50 unless changed: set it, together with the program's own power slider, so that the K4's ALC just shows while transmitting. A program that uses TCI for control alone, with its audio through a sound card, transmits as it would through CAT.

Only programs on the same computer can connect, as for CAT, since anything that can connect can put the K4 in transmit. A program cannot turn the K4 off through TCI.

Debugging

No guarantee is made that this software will run on a particular machine or distribution. However, debugging information may be useful to the developer. Problems can be reported by opening an issue at *https://github.com/DaleFarnsworth/K4-Companion/issues*. To initiate debugging, use the '-d' option: *./k4companion -d .* This will show a running list of actions by the script onscreen. To save the debugging information to a file, use *./k4companion -d > mydebugfile 2>&1.* The debug file can be named anything you like and the file will be created if it doesn't exist already. To append an existing log file, use *./k4companion -d >> mydebugfile 2>&1.* The file can then be posted or e-mailed to Dale, W7DA. Information from dmesg may be helpful as well: *sudo dmesg ,* although more likely if there is a system failure somewhere and not necessarily with K4 Companion.

To include the keyer's timing, which helps with a keying problem, add --da keyer: ./k4companion --da keyer >> mydebugfile 2>&1.

To include the audio's timing, which helps when the received audio is choppy or breaks up, add --da audio: ./k4companion --da audio >> mydebugfile 2>&1.

To include what the K-Pod reports, add --da kpod. When no K-Pod is found, it also lists every HID (USB input) device K4 Companion can see, with its vendor and product IDs, which shows whether a K-Pod that is plugged in is visible to it, perhaps under other IDs, or not at all, as when Windows has given it another driver. To include what CAT programs send through port 9200 and what they are answered, add --da cat. To include every MIDI message the keying device sends, and what is sent to a Vail adapter, add --da midi. To include what TCI programs send and are sent, add --da tci.

On Windows, k4companion.exe has no console, so its debugging information cannot be seen. Use k4companion-debug.exe instead: the same program with a console, taking the same options. In a Command Prompt, in the folder holding it, use *k4companion-debug -d > mydebugfile 2>&1*, adding *--da keyer* or *--da audio* as above if needed. The installer puts it in the installation folder, beside k4companion.exe. In any case, report the problem in as much detail as possible, and copy any error messages that do appear.

Notes on Operation

Audio functions on Linux can sometimes be challenging. A working knowledge of ALSA, alsamixer, and alsactl is useful. Pavucontrol can assist with selecting the correct audio input and output. Be sure to check the Configuration tab in pavucontrol, and if audio is not working, try other combinations. These may include “Stereo Duplex,” “Play HiFi quality Music,” and “Pro Audio,” among others.

Regarding Windows: I have noted it is prone to random audio failures, including disabling the audio for unknown reasons. You might have to dig into the subsystem a bit. I recommend using the Sound App, not the control panel. Use *mmsys.cpl* to access the Sound App. Control panel is often not very enlightening and the Sound App gives more information. As always, YMMV − your mileage may vary!

An experiment proved that K4 Companion will run on WSL – Windows Subsystem for Linux, but it is not easy to set up, and there are quite a few stumbling blocks. Choppy audio is probably the greatest impediment, probably due to adding extra layers of complexity to the audio subsystems. It does run under Parallels in macOS, as well as other emulators. Significant improvements have been made to the audio subsystem of K4 Companion, but there are simply limits to the way various operating systems handle sound reproduction.

Enjoy!

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Recent Revision History

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v2.2:
    Add a TCI server, for digital-mode programs
       Control, receive audio and transmit audio on port 40001.

    Base the panadapter on the K4's REF, with Ref and Scale dropdowns

    Place the waterfall's colours by the noise floor

    Scale the mini-pan from its averaged trace, without jumping

    Widen editable dropdowns enough for their widest value

    Support audio devices at 96 kHz

v2.1.3:
    Fix the FM Deviation menu entries, now shown with sliders

    Show ADC Overflow as a warning that doesn't stop the app

    Have Power off no longer report the connection as lost

v2.1.2:
    Key a Vail adapter's paddles with K4 Companion's keyer in Passthrough

    Show a menu value the K4 gives as words, such as Ext Ref lock

v2.1.1:
    Add a Frequency dropdown to the SSTV window's Transmit tab

    List the HID devices seen when no K-Pod is found, with --da kpod

    Say on the K-Pod tab that the K-Pod goes into this computer

    Fix toggling Auto Char Space, and the auto spacing defaults

v2.1:
    Add an SSTV window, receiving and sending pictures
       Martin, Scottie, Robot and PD modes; Autotune; pictures saved
       as they come in.

    Add an RTTY window
       Sent by the K4 in FSK D and by K4 Companion in AFSK A; AFC,
       Autotune and Search; a tuning strip; decode threshold,
       squelch and the K4's Dual-Tone filter.

    Add macro tags and call capture to the CW and RTTY windows

    Add a Spot tone to the CW window

    Rename SSNR's R selection C, have it turn SSNR on, and add AC

    Draw the waterfall in yellow on a dark blue background

    Add an Autoscale (AS) button to the panadapter window
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Remember the panadapter's Scale for each band

Add a WF setting choosing which signals the waterfall shows

Add support for the Elecraft K-Pod

Add installers for Linux

Offer the installers and manuals with each GitHub release

Let other programs share K4 Companion's connection on port 9200

Offer None as the audio input and output device

Support audio devices at 44.1 kHz

Show RTTY's mark and space tones on the panadapter

Restore each window on the monitor it was left on

Log MIDI messages with --da midi

Add a CMP dropdown to the Main tab

Turn compression off when SSTV or RTTY is picked

v2.0:

Rebuild the user interface on Qt (PyQt6)

Add a panadapter and waterfall window for each receiver

Add a keyer for paddles or a straight key connected to the computer
MoMIDI, serial and Vail adapter devices; Plain Iambic, Iambic A, Iambic B and Ultimatic modes; weighting; automatic character and word spacing.

Generate the sidetone locally when keying from paddles or a key
The K4's monitor sidetone is muted while keying.

Connect to the K4 on port 9204, encrypted with TLS, by default

Move the NB, SSNR and LMSNR controls to the Main tab

Add an SSNR reset selection and a 1 kHz bandwidth selection

Add exclusive mode for audio devices on Windows

Remove the Receive Latency setting

Replace the CW window's More tab with a Keying tab
The More tab's controls are now on the CW tab.

Split Autospot into separate Main and Sub buttons

Say why a connection was lost, and try to connect again

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v1.6.2:
    Add support for setting SSNR noise reduction

v1.5.8:
    Fix AF Center Pitch "Normal" button
       Clicking the button would cause the associated slider to briefly go
       to its maximum value before going to the "Normal" value. Now it doesn't
       go to the maximum value.

    Properly show/hide CW window on split mode change

    Fix the release version in the FW Versions list

    Change the Transmit button color to match the VFO

    Make audio device settings apply to all profiles
       Done so the user only has to set them once, not for each profile.

v1.5.7:
    Add hidden reverse_mouse wheel direction option
```